Parthendhar D

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.72
Tirumala Junior College	Class XII (Board of Intermediate Education Andhra Pradesh)	2020	92.30%
The Sun School	Class X (CBSE)	2018	89.50%

EXPERIENCE

Raam4wheelers LLP (MG Motors)

Technical Trainee- Consultant | *[April, 2024] - [July, 2024]*

- Gained hands-on exposure to **automotive service operations**, including vehicle diagnostics, maintenance workflows, and dealership management.
- Assisted in optimizing **vehicle logistics, workflow coordination, and traffic management** within the service center to improve operational efficiency.

SKILLS

CAE: Ansys Workbench, ABAQUS, COMSOL, Hypermesh

CAD: SolidWorks, AutoCAD, Catia

Programming Languages: Python (NumPy, Pandas, PyTorch, Scikit-Learn, SciPy), MATLAB, Fortran

Machine Learning: Regression Models, Random Forest, ANN, Deep Neural Networks, CNN, RNN

Problem solving, Teamwork, Communication skills.

RELEVANT COURSES

Finite Element Method, Machine Learning, Strength Of Materials, Advanced Mechanics Of Solids, Fracture Mechanics,

Vibration and Dynamics, Advanced Finite Element Method, Design of Machine Elements, Thermodynamics, Fluid Mechanics

MTECH THESIS

Numerical Framework for Homogenization of Thermo-Mechanical Properties of SiC-SiC Woven Composites and Structural Analysis of Ceramic-Based Divergent Slave Petal for Aero-Engine Applications *[August 2026-]*

- Developed a **Finite Element (FE) framework for numerical homogenization** of SiC-SiC woven composite RVE to predict effective thermo-mechanical properties.
- Investigated the influence of **woven architecture and fiber arrangement** on the effective material response.
- Utilized **Abaqus and Python scripting** to extract volume-averaged responses and compute effective thermo-mechanical properties.
- Developing a **Finite Element Model of the ceramic-based divergent slave petal assembly** for thermo-mechanical structural analysis under operating conditions.

PROJECTS

Thermo-Mechanical Finite Element Analysis of C/C Composite Multi-Disc Brakes for High-Speed Aircraft

[April 2026 - July 2026]

- Developed a **finite element framework in Abaqus** to simulate the thermo-mechanical response and wear behaviour of multi-disc C/C composite braking systems.
- Performed **coupled thermo-mechanical analysis** to investigate temperature evolution and structural response under successive braking conditions.
- Automated sequential braking simulations and accumulated wear calculations using **Python scripting**, enabling efficient analysis of multiple braking events.

Data-Driven Surrogate Modeling for Fracture Mechanics using Python, Abaqus API, and Machine Learning:

- Developed a **Finite Element Analysis (FEA) model** to evaluate the **Mode I Stress Intensity Factor (SIF)** in single-edge cracked plates.
- Automated **parametric simulations** using **Python script** to investigate the influence of crack length on fracture response.
- Developed regression-based predictive model to estimate SIFs from simulation data, with 99% Accuracy.

Machine Learning-Based Material Selection for Electric Vehicle Chassis

- Developed a **multi-layer Artificial Neural Network (ANN)**-based predictive framework.
- Implemented **data preprocessing, feature engineering, model development and hyperparameter tuning.**
- Achieved a reported **R^2 score of 0.9846** on the evaluated dataset.

Graph Neural Network for Machining Feature Recognition in CAD B-Rep Models:

- Built a **GNN** over **B-Rep face-adjacency graphs** to **classify machining features** in CAD models (MFCAD++, 25 classes).
- Implemented an end-to-end pipeline with automated **feature encoding**, relational graph convolution.
- Applied relational graph convolution with per-edge-type message passing; **achieved 90% face accuracy**.

HOBBIES AND INTERESTS

Automobiles, Cricket, Geopolitics, Movies, Food, F1, Sim-racing